Chapter 8: Two – Level Fractional Factorial Design

As the **number** of **factors** in a2^k factorial design **increases**, the **number** of **runs** required for a complete replicate of the design **rapidly outgrows** the resources of most experimenters. In fact, full 2^k factorial designs are seldom used in practice for large k (say $k \geq 7$).

If the experimenter can reasonably **assume** that certain **high-order interactions** are **negligible**, information on the **main effects** and **low-order interactions** may be obtained by running only a fraction of the complete factorial experiment.

For economic reasons, fractional factorial designs, which consist of a subset or **fraction** of full factorial designs, are commonly used.

We will discuss about **fractional factorial design** in this chapter.

Consider a 2^6 factorial design. Even without replicates, there are $2^6 = 64$ observations required to estimate the effects.

In this design, there are: **6 main effects**, **15 two factor interactions**, 20 three- factor interactions, 15 four-factor interactions , 6 five-factor interactions and 1 six-factor interaction.

In this design only 6 of the 63 degrees of freedom correspond to the main effects, and only 15 degrees of freedom correspond to two-factor interactions. Therefore, there are only 21 degrees of freedom associated with effects that are likely to be major interest. The remaining 42 degrees of freedom are associated with three-factor and higher interactions. If three (or more) factor interactions are **not** of interest, then **fewer** observation are needed top estimate other effects.

A 2^{k-1} design, or "**one - half fraction of the 2^k design**" is one in which only half of the 2^k treatment combinations are observed.

A major use of fractional factorial designs is in **screening experiments**.

These are experiments in which **many factors** are considered with the purpose of identifying those factors (if any) that have large effects.

Screening experiments are usually performed in the **early stages of a project** when it is likely that many of the factors initially considered have little or no effect on the response.

The factors that are identified as important are then investigated more thoroughly in **subsequent** experiments.

"**Optimal**" fractions are chosen according to the **resolution** and **minimum aberration criteria**.

Recall that in a 2^3 design, with 4 runs in each of 2 blocks, we confound ABC with blocks in the following way:

Treatment Combinations	Factorial Effect								Block
	I	A	B	AB	C	AC	BC	ABC	
(1)	+	−	−	+	−	+	+	−	1
a	+	+	−	−	−	−	+	+	2
b	+	−	+	−	−	+	−	+	2
ab	+	+	+	+	−	−	−	−	1
c	+	−	−	+	+	−	−	+	2
ac	+	+	−	−	+	+	−	−	1
bc	+	−	+	−	+	−	+	−	1
abc	+	+	+	+	+	+	+	+	2

One-Half Fraction of Factorial Designs:

Consider a situation in which **three factors**, each at **two levels**, are of interest, but the experimenters **cannot** afford to run all $2^3 = 8$ treatment combinations. They can, however, afford **four** runs. This suggests a **one-half fraction** of a design.

Suppose we select the four treatment combinations $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ and $\boldsymbol{abc}$, as our **one-half** fraction. These runs are shown in the **top half** of the following table.

Treatment Combinations	Factorial Effect							
	I	A	B	AB	C	AC	BC	ABC
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
c	+	−	−	+	+	−	−	+
abc	+	+	+	+	+	+	+	+
ab	+	+	+	+	−	−	−	−
ac	+	+	−	−	+	+	−	−
bc	+	−	+	−	+	−	+	−
(1)	+	−	−	+	−	+	+	−

Because the design contains $2^{3-1} = 4$ treatment combinations, a one-half fraction of the 2^3 design is often called a 2^{3-1} design.

Treatment Combinations	Factorial Effect							
	I	A	B	AB	C	AC	BC	ABC
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
c	+	−	−	+	+	−	−	+
abc	+	+	+	+	+	+	+	+
ab	+	+	+	+	−	−	−	−
ac	+	+	−	−	+	+	−	−
bc	+	−	+	−	+	−	+	−
(1)	+	−	−	+	−	+	+	−

If we run only top half, then the design uses $\{a, b, c, abc\}$.

These are those for which $\boldsymbol{ABC} = +$; since also $\boldsymbol{I} = +$, we say that the "**defining relation**" for the design is $\boldsymbol{ABC} = \boldsymbol{I}$ and we refer to the "**word**" **ABC** as the "**generator**'" of the design.

If we used only those combinations with $\boldsymbol{A} = +$, then $\boldsymbol{A} = \boldsymbol{I}$ would be the defining relation and $\boldsymbol{A}$ the generating word.

Treatment Combinations	Factorial Effect							
	I	A	B	AB	C	AC	BC	ABC
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
c	+	−	−	+	+	−	−	+
abc	+	+	+	+	+	+	+	+
ab	+	+	+	+	−	−	−	−
ac	+	+	−	−	+	+	−	−
bc	+	−	+	−	+	−	+	−
(1)	+	−	−	+	−	+	+	−

The one-half fraction with defining relation $ABC = I$ is called the "**principal fraction**".

The other half, in which $ABC = -$, is called the "**alternate**" or "**complementary**" fraction, and has defining relation $-ABC = I$.

In general, the **defining relation** of a fractional factorial design is always the set of **all columns** that are equal to identity column I.

If we uses $\{a, b, c, abc\}$, our one-half fraction is:

Treatment Combinations	Factorial Effect							
	I	A	B	AB	C	AC	BC	ABC
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
c	+	−	−	+	+	−	−	+
abc	+	+	+	+	+	+	+	+

The estimates of the effects are obtained by applying the ± signs appropriately. We use [·]'s to distinguish these from the full factorial estimates.

$$[A] = \frac{a - b - c + abc}{2} = [\mathrm{BC}]$$

$$[B] = \frac{-a + b - c + abc}{2} = [\mathrm{AC}]$$

$$[C] = \frac{-a - b + c + abc}{2} = [\mathrm{AB}]$$

If we uses $\{a, b, c, abc\}$, our one-half fraction is:

$$[A] = \frac{a - b - c + abc}{2} = [\mathrm{BC}]$$
$$[B] = \frac{-a + b - c + abc}{2} = [\mathrm{AC}]$$
$$[C] = \frac{-a - b + c + abc}{2} = [\mathrm{AB}]$$

Therefore, it is impossible to difference between A and BC, B and AC, C and AB. Recall that in a full factorial design, we have

$$A = \frac{1}{4}\left[(a-(1)) + (ab-b) + (ac-c) + (abc-bc)\right]$$
$$\mathrm{BC} = \frac{1}{2}\left\{\frac{\left[(abc-ac)+(bc-c)\right]}{2} - \frac{\left[(ab-a)+(b-(1))\right]}{2}\right\}$$

Therefore, $A + BC = \frac{a-b-c+abc}{2}$. In fact, when we estimate A, B and C, we are really estimating $A + BC$, $B + AC$, and $C + AB$.

If we uses $\{a, b, c, abc\}$, our one-half fraction is:

$$[A] = \frac{a - b - c + abc}{2} = [\mathrm{BC}]$$
$$[B] = \frac{-a + b - c + abc}{2} = [\mathrm{AC}]$$
$$[C] = \frac{-a - b + c + abc}{2} = [\mathrm{AB}]$$

Therefore, it is impossible to difference between A and BC, B and AC, and C and AB. In fact, when we estimate A, B and C, we are really estimating $A + BC$, $B + AC$, and $C + AB$. Two or more effects that have this property are called **aliases**.

In our example, A and BC are aliases, B and AC are aliases, and C and AB.We indicate this by the following notation:

$$[\boldsymbol{A}] \rightarrow \boldsymbol{A} + \boldsymbol{BC}, \qquad [\boldsymbol{B}] \rightarrow \boldsymbol{B} + \boldsymbol{AC}, \qquad [\boldsymbol{C}] \rightarrow \boldsymbol{C} + \boldsymbol{AB}$$

If we uses $\{a, b, c, abc\}$, our one-half fraction is:

$$[A] = \frac{a - b - c + abc}{2} = [\mathrm{BC}]$$
$$[B] = \frac{-a + b - c + abc}{2} = [\mathrm{AC}]$$
$$[C] = \frac{-a - b + c + abc}{2} = [\mathrm{AB}]$$

The alias structure for this design may be easily determined by using the defining relation $\boldsymbol{ABC = I}$ and multiplication (mode 2).

$$I = ABC \rightarrow A = A^2BC = BC$$

$$I = ABC \rightarrow B = AB^2C = AC$$

$$I = ABC \rightarrow C = ABC^2 = AB$$

Treatment Combinations	Factorial Effect							
	I	A	B	AB	C	AC	BC	ABC
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
c	+	−	−	+	+	−	−	+
abc	+	+	+	+	+	+	+	+
ab	+	+	+	+	−	−	−	−
ac	+	+	−	−	+	+	−	−
bc	+	−	+	−	+	−	+	−
(1)	+	−	−	+	−	+	+	−

The one-half fraction with defining relation $ABC = I$ is called the "**principal fraction**".

The other half, n which $ABC = -$, is called the "**alternate**" or "**complementary**" fraction, and has defining relation $-ABC = I$.

If we uses $\{ab, ac, bc, (1)\}$, our one-half fraction is:

Treatment	Factorial Effect							
Combinations	I	A	B	AB	C	AC	BC	ABC
ab	+	+	+	+	−	−	−	−
ac	+	+	−	−	+	+	−	−
bc	+	−	+	−	+	−	+	−
(1)	+	−	−	+	−	+	+	−

The estimates from **complementary fraction** are:

$$[A]' = \frac{ab + ac - bc - (1)}{2} = -[\mathrm{BC}]'$$

$$[B]' = \frac{ab - ac + bc - (1)}{2} = -[\mathrm{AC}]'$$

$$[C]' = \frac{-ab + ac + bc - (1)}{2} = -[\mathrm{AB}]'$$

Similarly, in a full factorial design, $A - BC = \frac{ab-ac-bc+(1)}{2}$. Therefore,

$$[\boldsymbol{A}]' \rightarrow \boldsymbol{A} - \boldsymbol{BC}, \quad [\boldsymbol{B}]' \rightarrow \boldsymbol{B} - \boldsymbol{AC}, \quad [\boldsymbol{C}]' \rightarrow \boldsymbol{C} - \boldsymbol{AB}$$

If we uses $\{a, b, c, abc\}$, our one-half fraction is:

$$[A]' = \frac{ab - ac - bc + (1)}{2} = -[\mathrm{BC}]'$$
$$[B]' = \frac{ab - ac + bc - (1)}{2} = -[\mathrm{AC}]'$$
$$[C]' = \frac{-ab + ac + bc - (1)}{2} = -[\mathrm{AB}]'$$

The alias structure for this design may be easily determined by using the defining relation $\boldsymbol{-ABC = I}$ and multiplication (mode 2).

$$I = -ABC \rightarrow A = -A^2BC = -BC$$

$$I = -ABC \rightarrow B = -AB^2C = -AC$$

$$I = -ABC \rightarrow C = -ABC^2 = -AB$$

If we were to **run one-half fraction**, and then decide to **run the other**, we could view the entire experiment as a full 2^3 factorial run in two blocks, with ABC confounded with blocks.

We could combine our estimates to get estimates of all main effects and two-factor interactions:

$$[A] = \frac{a - b - c + abc}{2} = [\mathrm{BC}]$$

$$[A]' = \frac{ab - ac - bc + (1)}{2} = -[\mathrm{BC}]'$$

Therefore, for example, we have

$$\frac{[A] + [A]'}{2} = \frac{a - b - c + abc}{4} + \frac{ab - ac - bc + (1)}{4} = A$$

$$\frac{[A] - [A]'}{2} = \frac{a - b - c + abc}{4} - \frac{ab - ac - bc + (1)}{4} = BC$$

We probably wouldn't want to run one half of a 2^3 factorial unless we were confident that the two factor interactions were not significant.

Example 1: Consider an experiment to improve a heat treatment process on ruck leaf springs (Pignatiello, J. J., Jr., and Ramberg, J. S. (1985), Discussion of "*Off-line quality control, parameter design, and the Taguchi method*" by Kackar, R. N., Journal of Quality Technology, **17**, 198-206).

The heat treatment that forms the camber (or curvature) in leaf spring consists of heating in a high-temperature furnace, processing by a forming machine, and quenching in an oil bath.

The height of an unloaded spring, known as free height, is an important quality characteristic whose **target value is 8 inches**. An experimental **goal** is to **make the variation about the target 8.0 as small as possible**.

Five factors, listed in the next page table, were chosen across the various stages of the process.

Example 1: Five factors were chosen across the various stages of the process.

Factor	Level	
	−	+
A: High heat Temperature (^{o}F)	1840	1880
B: Heating time (seconds)	23	25
C: Transfer time (seconds)	10	12
D: Hold-down time (seconds)	2	3
E: Quench oil temperature (^{o}F)	130-150	150-170

Heating time are from heating stage,

Transfer time is the time it takes the conveyer to transport the spring from the furnace to the forming machine.

Hold-down time is the time that the springs are held in a high-pressure press to form the camber.

Quench oil temperature is from the quenching stage.

Example 1 (Cont.): Five factors were chosen across the various stages of the process.

Factor	Level	
	$-$	$+$
A: High heat Temperature (^{o}F)	1840	1880
B: Heating time (seconds)	23	25
C: Transfer time (seconds)	10	12
D: Hold-down time (seconds)	2	3
E: Quench oil temperature (^{o}F)	130-150	150-170

The five factors are all **quantitative**, but levels of E do not have fixed values. Because the quench oil temperature could not be controlled precisely, the experimenters could only give a range of values for each level.

We treat the rages 130-150 ^{o}F and 150-170 ^{o}F as the low and high levels of factor E.

Example 1 (Cont.): The experimental design that was used was a half fraction of a 2^5 full factorial design with each run being replicated three times. The design matrix and free height data are given in the following table:

Factor					Free Height					
A	B	C	D	E				$\bar{y}_i$	s_i^2	Lns_i^2
−	+	+	−	−	7.78	7.78	7.81	7.7900	0.0003	−8.1117
+	+	+	+	−	8.15	8.18	7.88	8.0700	0.0273	−3.6009
−	−	+	+	−	7.50	7.56	7.50	7.5200	0.0012	−6.7254
+	−	+	−	−	7.59	7.56	7.75	7.6333	0.0104	−4.5627
−	+	−	+	−	7.94	8.00	7.88	7.9400	0.0034	−5.6268
+	+	−	−	−	7.69	8.09	8.06	7.9467	0.0496	−3.0031
−	−	−	−	−	7.56	7.62	7.44	7.5400	0.0084	−4.7795
+	−	−	+	−	7.56	7.81	7.69	7.6867	0.0156	−4.1583
−	+	+	−	+	7.50	7.25	7.12	7.2900	0.0373	−3.2888
+	+	+	+	+	7.88	7.88	7.44	7.7333	0.0645	−2.7406
−	−	+	+	+	7.50	7.56	7.50	7.5200	0.0012	−6.7254
+	−	+	−	+	7.63	7.75	7.56	7.6467	0.0092	−4.6849
−	+	−	+	+	7.32	7.44	7.44	7.4000	0.0048	−5.3391
+	+	−	−	+	7.56	7.69	7.62	7.6233	0.0042	−5.4648
−	−	−	−	+	7.18	7.18	7.25	7.2033	0.0016	−6.4171
+	−	−	+	+	7.81	7.50	7.59	7.6333	0.0254	−3.6717

Example 1 (Cont.): The experimental design that was used was a half fraction of a 2^5 full factorial design with each run being replicated three times.

For this experiment, factor $\boldsymbol{D}$ is assigned to the column that equals the **product** of the columns for factors $\boldsymbol{A}, \boldsymbol{B}$, and $\boldsymbol{C}$; that is each entry in the column for D is the same as the product of the corresponding entries in the columns for A, B, and C. Therefore, we have

$$\mathbf{D} = \boldsymbol{ABC} \text{ or } \mathbf{ABCD} = \boldsymbol{I}$$

Because the $\mathbf{2^{5-1}}$ **design** has only **15 degrees of freedom** for estimating factorial effects, it **cannot estimate all 31 factorial effects** among the factors A, B, C, D and E.

Aliasing of effects is a price one must pay for choosing a smaller design.

Example 1 (Cont.): The experimental design that was used was a half fraction of a 2^5 full factorial design with **ABCD**$= \boldsymbol{I}$ and each run being replicated three times.

Aliasing of effects is a price one must pay for choosing a smaller design.

By multiplying both sides of **ABCD**$= \boldsymbol{I}$ all **15 aliasing** relationship can be obtained.

$A = BCD,$ $B = ACD,$ $C = ABD,$ $D = ABC,$

$\mathrm{AB} = CD,$ $AC = BD,$ $AD = BC,$

$E = ABCDE,$ $AE = BCDE,$ $BE = ACDE,$ $CE = ABDE,$
$DE = ABCE,$

$ABE = CDE,$ $ACE = BDE,$ $ADE = BCE$

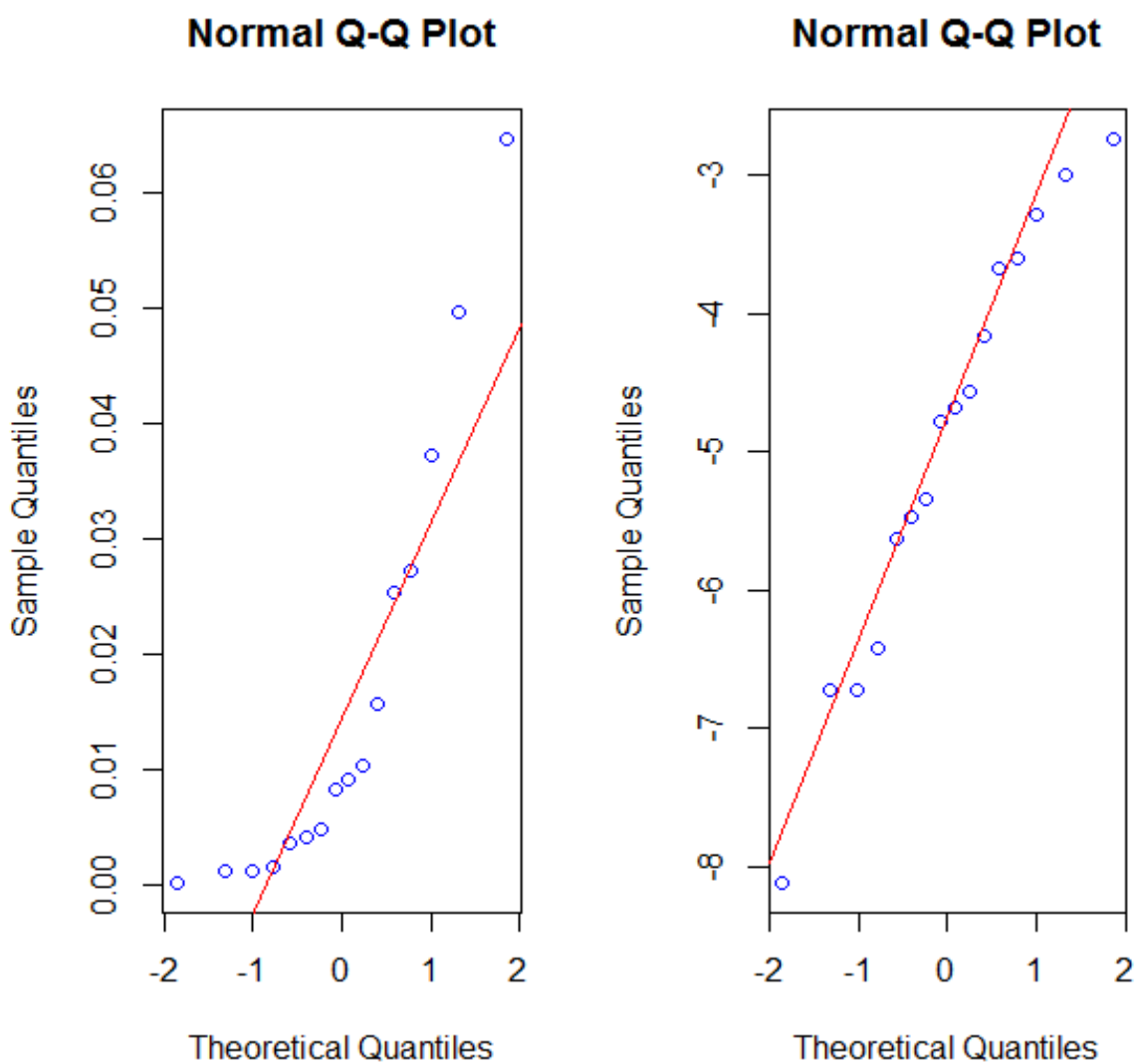
R Codes to Enter Data:

```
> Height = c(7.78,8.15,7.50,7.59,7.94,7.69,7.56,7.56,
        7.50,7.88,7.50,7.63,7.32,7.56,7.18,7.81,
        7.78,8.18,7.56,7.56,8.00,8.09,7.62,7.81,
        7.25,7.88,7.56,7.75,7.44,7.69,7.18,7.50,
        7.81,7.88,7.50,7.75,7.88,8.06,7.44,7.69,
        7.12,7.44,7.50,7.56,7.44,7.62,7.25,7.59)

> data=matrix(Height,nrow=16)
>colnames(data)=c("Rep I", "Rep II", "Rep III")
> data
```

	Rep I	Rep II	Rep III
[1,]	7.78	7.78	7.81
[2,]	8.15	8.18	7.88
[3,]	7.50	7.56	7.50
[4,]	7.59	7.56	7.75
[5,]	7.94	8.00	7.88
[6,]	7.69	8.09	8.06
[7,]	7.56	7.62	7.44
[8,]	7.56	7.81	7.69
[9,]	7.50	7.25	7.12
[10,]	7.88	7.88	7.44
[11,]	7.50	7.56	7.50
[12,]	7.63	7.75	7.56
[13,]	7.32	7.44	7.44
[14,]	7.56	7.69	7.62
[15,]	7.18	7.18	7.25
[16,]	7.81	7.50	7.59

R Codes to Obtain Normal Plots:

```
> s2=apply(data,1,var)

> lns2=log(apply(data,1,var))


> windows()
> par(mfrow=c(1,2))

> qqnorm(s2,col="blue")
> qqline(s2, col=2)

> qqnorm(lns2,col="blue")
> qqline(lns2, col=2)
```



R Codes to Enter Data:

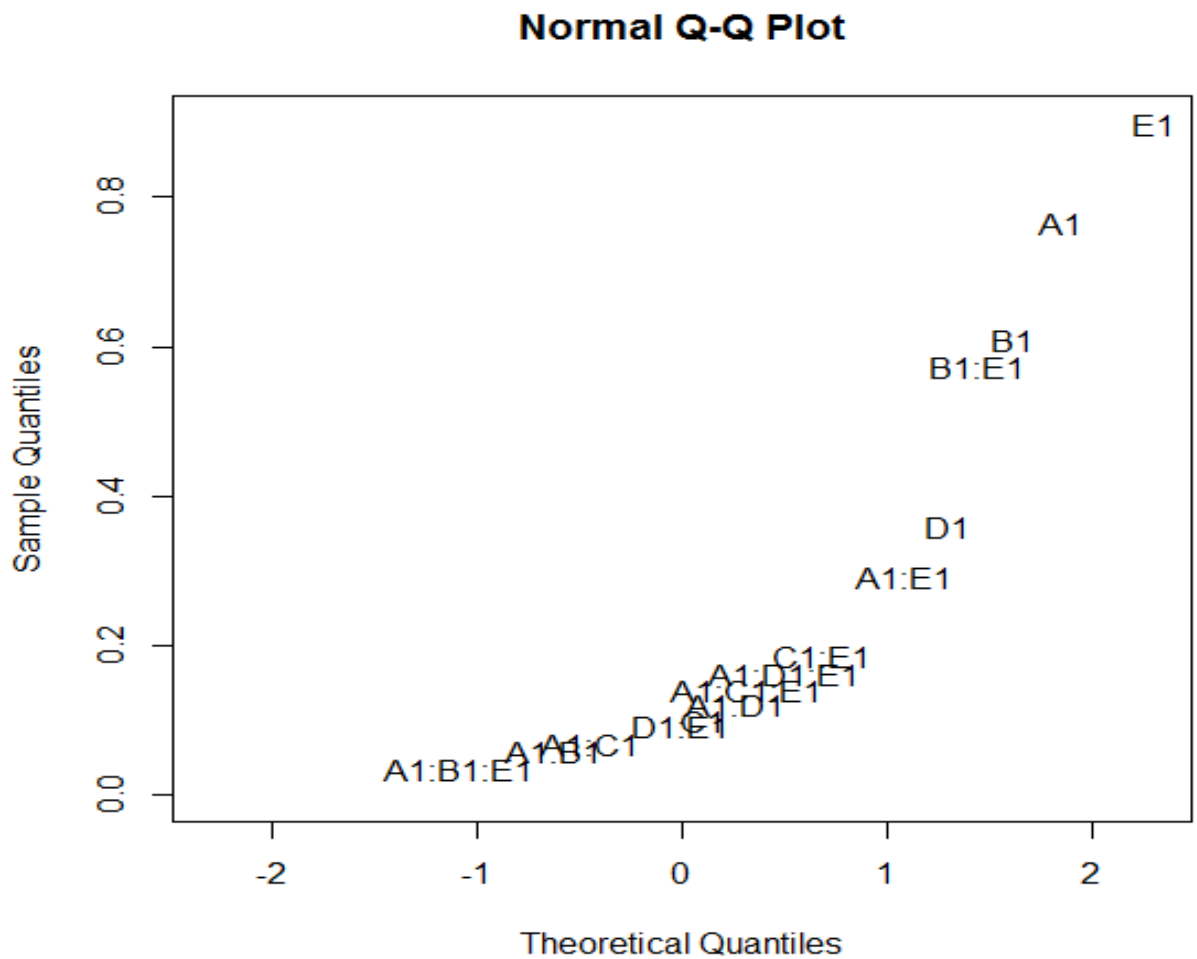
```
> A = rep(c(-1,1), times = 8)
> B = rep(c(1,-1), each=2,times = 4)
> C = rep(c(1,-1), each=4,times = 2)

> D = A*B*C

> A = as.factor(A)
> B = as.factor(B)
> C = as.factor(C)
> D = as.factor(D)
> E=factor(rep(c(-1,1), each= 8))

> data = data.frame(A, B, C, D, E, data)
> data
```

	A	B	C	D	E	Rep.I	Rep.II	Rep.III
1	-1	1	1	-1	-1	7.78	7.78	7.81
2	1	1	1	1	-1	8.15	8.18	7.88
3	-1	-1	1	1	-1	7.50	7.56	7.50
4	1	-1	1	-1	-1	7.59	7.56	7.75
5	-1	1	-1	1	-1	7.94	8.00	7.88
6	1	1	-1	-1	-1	7.69	8.09	8.06
7	-1	-1	-1	-1	-1	7.56	7.62	7.44
8	1	-1	-1	1	-1	7.56	7.81	7.69
9	-1	1	1	-1	1	7.50	7.25	7.12
10	1	1	1	1	1	7.88	7.88	7.44
11	-1	-1	1	1	1	7.50	7.56	7.50
12	1	-1	1	-1	1	7.63	7.75	7.56
13	-1	1	-1	1	1	7.32	7.44	7.44
14	1	1	-1	-1	1	7.56	7.69	7.62
15	-1	-1	-1	-1	1	7.18	7.18	7.25
16	1	-1	-1	1	1	7.81	7.50	7.59

Half-Normal plot:

```
> A = rep(A,3)
> B = rep(B,3)
> C = rep(C,3)
> D = rep(D,3)
> E = rep(E,3)
```

```
> model = lm(Height ~
      (A+B+C+D+E)^5)
```




```
> model$effects
> model.effects = abs(model$effects[-1])

> qq = qqnorm(model.effects, type="n")
> text(qq$x, qq$y, labels = names(model.effects))
```

R Code for ANOVA Table:

> anova(model)

```
Analysis of Variance Table

Response: Height
          Df  Sum Sq Mean Sq F value    Pr(>F)
A          1 0.58742 0.58742 35.4846 1.230e-06 ***
B          1 0.37277 0.37277 22.5181 4.160e-05 ***
C          1 0.00992 0.00992  0.5992  0.444577
D          1 0.12917 0.12917  7.8028  0.008735 **
E          1 0.80860 0.80860 48.8458 6.416e-08 ***
A:B        1 0.00350 0.00350  0.2116  0.648663
A:C        1 0.00460 0.00460  0.2780  0.601651
A:D        1 0.01505 0.01505  0.9093  0.347458
A:E        1 0.08585 0.08585  5.1861  0.029592 *
B:E        1 0.32835 0.32835 19.8350 9.645e-05 ***
C:E        1 0.03467 0.03467  2.0943  0.157579
D:E        1 0.00880 0.00880  0.5317  0.471191
A:B:E      1 0.00130 0.00130  0.0787  0.780932
A:C:E      1 0.01960 0.01960  1.1841  0.284647
A:D:E      1 0.02660 0.02660  1.6070  0.214065
Residuals 32 0.52973 0.01655
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

R Codes for Removing One Column:

```
> newdata = data[,-3]
> newdata
```

	A	B	D	E	Rep.I	Rep.II	Rep.III
1	-1	1	-1	-1	7.78	7.78	7.81
2	1	1	1	-1	8.15	8.18	7.88
3	-1	-1	1	-1	7.50	7.56	7.50
4	1	-1	-1	-1	7.59	7.56	7.75
5	-1	1	1	-1	7.94	8.00	7.88
6	1	1	-1	-1	7.69	8.09	8.06
7	-1	-1	-1	-1	7.56	7.62	7.44
8	1	-1	1	-1	7.56	7.81	7.69
9	-1	1	-1	1	7.50	7.25	7.12
10	1	1	1	1	7.88	7.88	7.44
11	-1	-1	1	1	7.50	7.56	7.50
12	1	-1	-1	1	7.63	7.75	7.56
13	-1	1	1	1	7.32	7.44	7.44
14	1	1	-1	1	7.56	7.69	7.62
15	-1	-1	-1	1	7.18	7.18	7.25
16	1	-1	1	1	7.81	7.50	7.59

R Codes for ANOVA Table:

```
> model2 = lm(Height ~A+B+D+E+B*E+A*E)
> anova(model)
```

Analysis of Variance Table

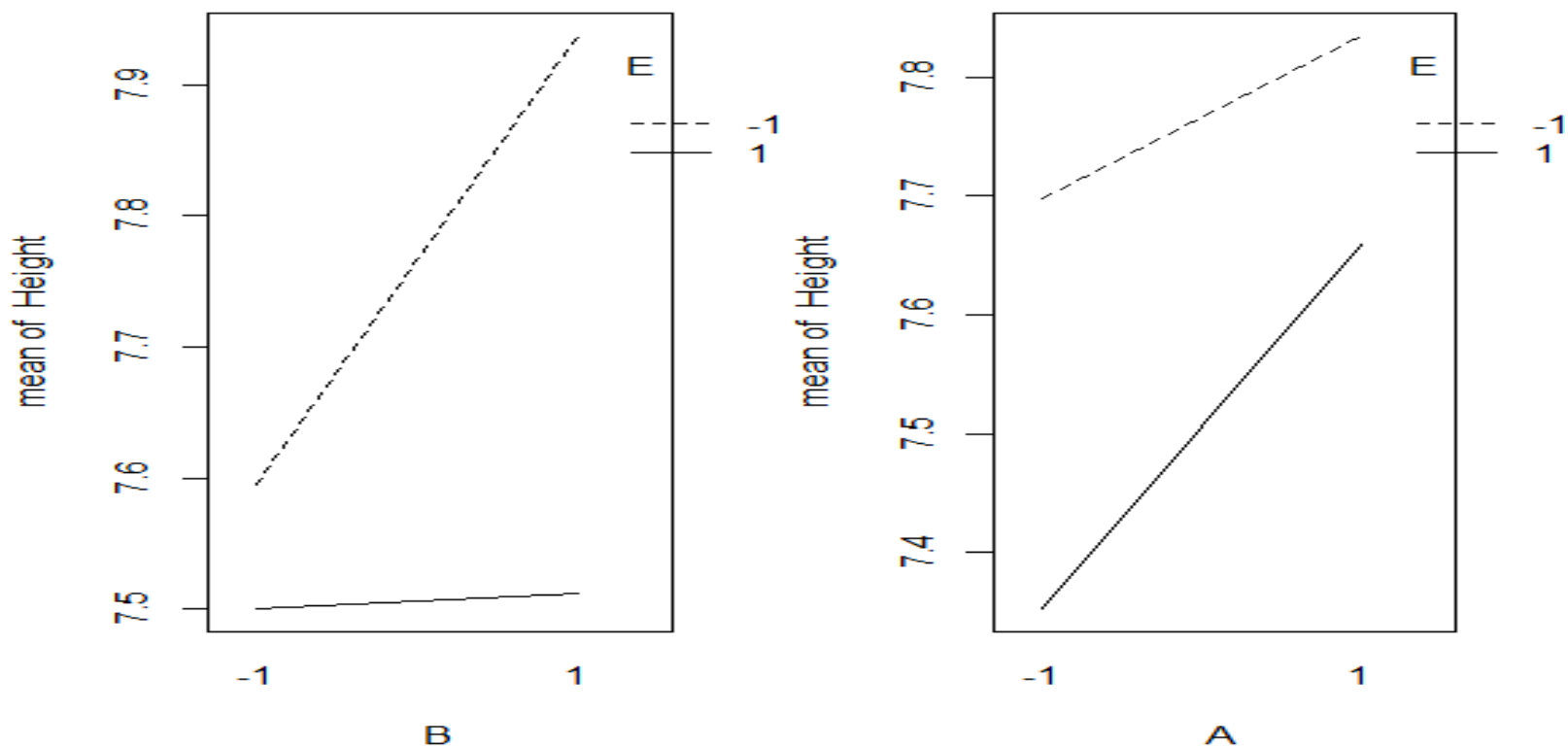
Response: Height

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
A	1	0.58742	0.58742	36.8380	3.446e-07 ***
B	1	0.37277	0.37277	23.3770	1.904e-05 ***
D	1	0.12917	0.12917	8.1004	0.006881 **
E	1	0.80860	0.80860	50.7088	1.113e-08 ***
B:E	1	0.32835	0.32835	20.5915	4.892e-05 ***
A:E	1	0.08585	0.08585	5.3839	0.025377 *
Residuals	41	0.65379	0.01595		

Signif. codes: 0 '*' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1**

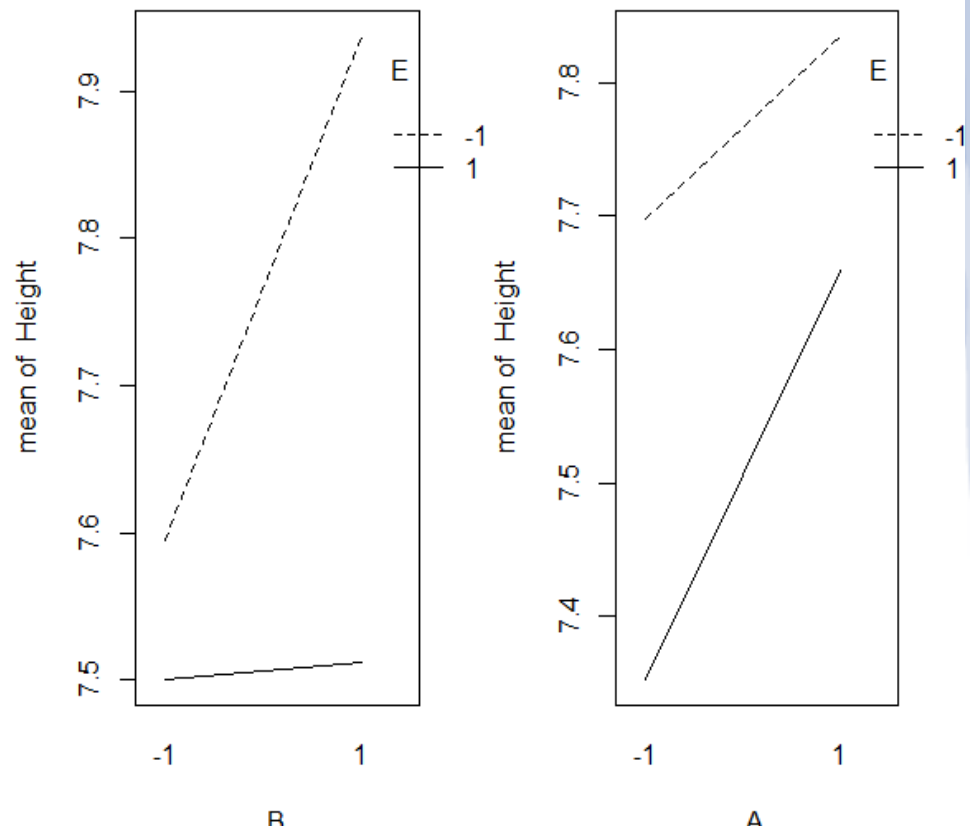
R Codes for Interaction Plots:

```
> windows()
> par(mfrow=c(1,2))
> interaction.plot(B,E,Height)
> interaction.plot(A,E,Height)
```



R Codes to Obtain Estimate of Regression Equation:

```
> A = rep(c(-1,1), times = 8)
> B = rep(c(1,-1), each=2,times = 4)
> C = rep(c(1,-1), each=4,times = 2)
> D = A*B*C
> E=  rep(c(-1,1), each= 8)

> A = rep(A,3)    > B = rep(B,3)
> D = rep(D,3)   > E = rep(E,3)

> model2= lm(Height ~
                A+B+D+E+B*E+A*E)
> model2
```




```
Call:
lm(formula = Height ~ A + B + D + E + B * E + A * E)

Coefficients:
(Intercept)            A            B            D            E          B:E
    7.63604      0.11063      0.08812      0.05188     -0.12979     -0.08271
        A:E
    0.04229
```

Example 1: To have **largest** estimate of height, we should choose

$$A = +, B = +, D = +, \text{and } E = -$$

Note that,

```
Call:
lm(formula = Height ~ A + B + D + E + B * E + A * E)

Coefficients:
(Intercept)            A            B            D            E          B:E
    7.63604      0.11063      0.08812      0.05188     -0.12979     -0.08271
        A:E
    0.04229
```

Therefore

$$\hat{y} = 7.63604 + 0.11063A + 0.08812B + 0.05188D - 0.12979E$$
$$- 0.08271B*E + 0.04229A*E$$

$$\hat{y} = 7.63604 + 0.11063(1) + 0.08812(1) + 0.05188(1) - 0.12979(-1)$$
$$- 0.08271(1)*(-1) + 0.04229(1)*(-1) = 8.05688$$

But we need to move estimate of height closer to the **target** value of 8.0 with **small variation**.

R Codes to Enter Data for L$n(s^2)$:

```
> A = rep(c(-1,1), times = 8)
> B = rep(c(1,-1), each=2,times = 4)
> C = rep(c(1,-1), each=4,times = 2)

> D = A*B*C

> A = as.factor(A)
> B = as.factor(B)
> C = as.factor(C)
> D = as.factor(D)
> E=factor(rep(c(-1,1), each= 8))

> data_lns2 = data.frame(A, B, C, D, E, lns2)
> data_lns2
```

	A	B	C	D	E	lns2
1	-1	1	1	-1	-1	-8.111728
2	1	1	1	1	-1	-3.600869
3	-1	-1	1	1	-1	-6.725434
4	1	-1	1	-1	-1	-4.562749
5	-1	1	-1	1	-1	-5.626821
6	1	1	-1	-1	-1	-3.003093
7	-1	-1	-1	-1	-1	-4.779524
8	1	-1	-1	1	-1	-4.158350
9	-1	1	1	-1	1	-3.288762
10	1	1	1	1	1	-2.740573
11	-1	-1	1	1	1	-6.725434
12	1	-1	1	-1	1	-4.684935
13	-1	1	-1	1	1	-5.339139
14	1	1	-1	-1	1	-5.464766
15	-1	-1	-1	-1	1	-6.417132
16	1	-1	-1	1	1	-3.671695

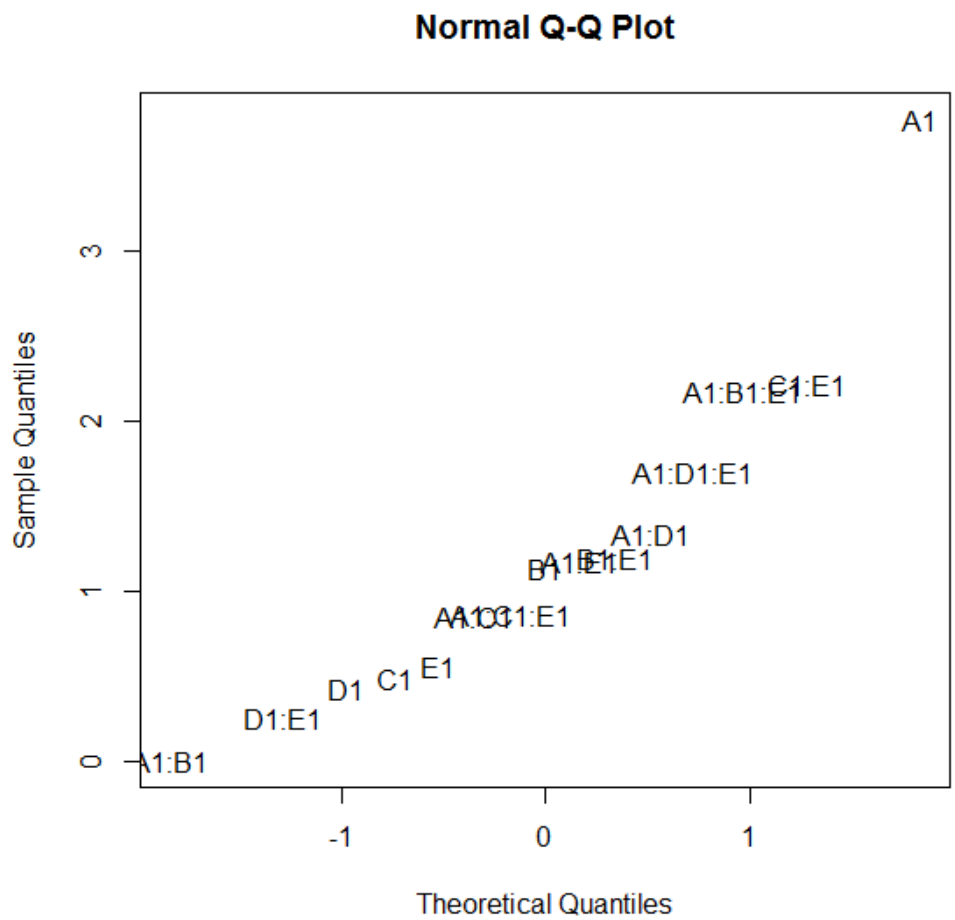
R Codes for ANOVA Table:

```
> model_lns2 = lm(lns2 ~ (A+B+C+D+E)^5)
> anova(model_lns2)
```

```
Analysis of Variance Table

Response: lns2
          Df  Sum Sq Mean Sq F value Pr(>F)
A          1 14.3015 14.3015
B          1  1.2936  1.2936
C          1  0.2450  0.2450
D          1  0.1858  0.1858
E          1  0.3125  0.3125
A:B        1  0.0000  0.0000
A:C        1  0.7214  0.7214
A:D        1  1.7981  1.7981
A:E        1  1.3865  1.3865
B:E        1  1.4295  1.4295
C:E        1  4.9351  4.9351
D:E        1  0.0667  0.0667
A:B:E      1  4.7460  4.7460
A:C:E      1  0.7481  0.7481
A:D:E      1  2.9143  2.9143
Residuals  0  0.0000
Warning message:
In anova.lm(model_lns2) :
  ANOVA F-tests on an essentially perfect fit are unreliable
```

Half-Normal plot:

```
> model_lns2 = lm(lns2 ~ (A+B+C+D+E)^5)
> anova(model_lns2)

> model_lns2$effects
> model_lns2.effects =
    abs(model_lns2$effects[-1])

> qq = qqnorm(model_lns2.effects,
              type="n")
> text(qq$x, qq$y,
       labels =
         names(model_lns2.effects))
```



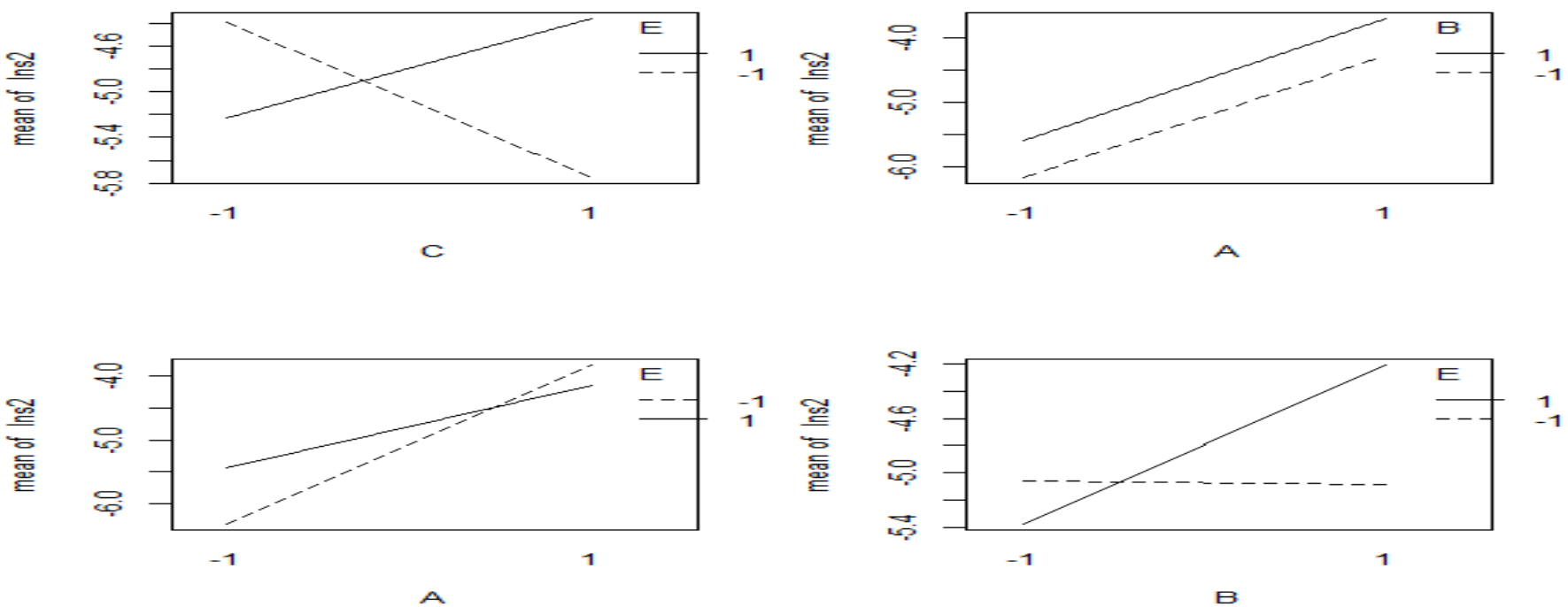
R Codes for ANOVA Table:

```
> model2_lns2 = lm(lns2 ~ A+C*E+A*B*E)
> anova(model2_lns2)
```

```
Analysis of Variance Table

Response: lns2
          Df  Sum Sq Mean Sq F value  Pr(>F)
A          1 14.3015 14.3015 13.3358 0.01068 *
C          1  0.2450  0.2450  0.2285 0.64958
E          1  0.3125  0.3125  0.2914 0.60875
B          1  1.2936  1.2936  1.2063 0.31418
C:E        1  4.9351  4.9351  4.6018 0.07561 .
A:B        1  0.0000  0.0000  0.0000 0.99766
A:E        1  1.3865  1.3865  1.2929 0.29888
E:B        1  1.4295  1.4295  1.3329 0.29219
A:E:B      1  4.7460  4.7460  4.4255 0.08007 .
Residuals  6  6.4345  1.0724
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

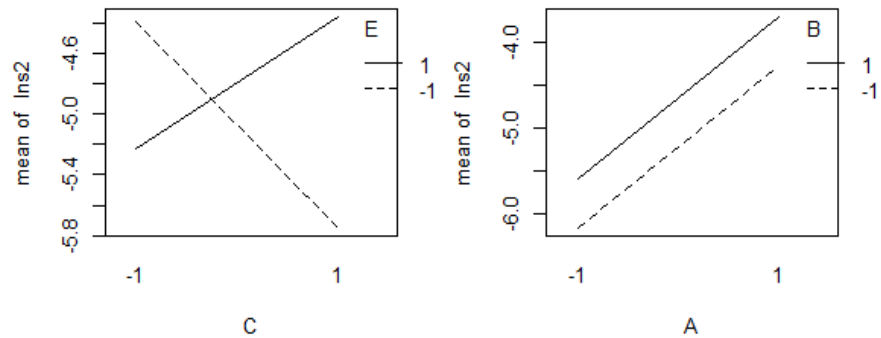
R Codes for Interaction Plots:

```
> windows()        > par(mfrow=c(2,2))
> interaction.plot(C,E,lns2)
> interaction.plot(A,B,lns2)
> interaction.plot(A,E,lns2)
> interaction.plot(B,E,lns2)
```

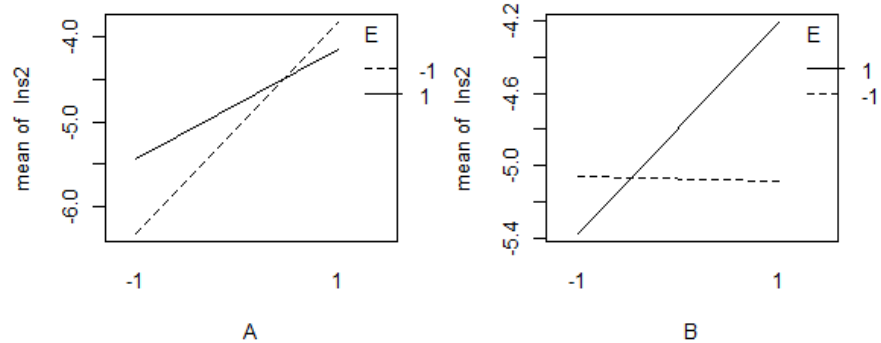


R Codes to Obtain Estimate of Regression Equation:

```
> A   = rep(c(-1,1), times = 8)
> B   = rep(c(1,-1), each=2,times = 4)
> C   = rep(c(1,-1), each=4,times = 2)
> E   = rep(c(-1,1), each= 8)
> CE  = C*E
> ABE = A*B*E
```




```
> model2_lns2 =
                lm(lns2 ~ A+CE+ABE)

> model2_lns2
```




```
Call:
lm(formula = lns2 ~ A + CE + ABE)

Coefficients:
(Intercept)            A           CE          ABE
    -4.9313       0.9454       0.5554      -0.5446
```

Example 1: To **reduce** the variation, we should choose

$$A = -, B = +, C = +, \text{and } E = -$$

Recall that,

> model2

Call:

lm(formula = Height ~ A + B + D + E + B * E + A * E)

Coefficients:

(Intercept)	A	B	D	E	B:E
7.63604	0.11063	0.08812	0.05188	-0.12979	-0.08271

A:E
0.04229

Therefore

$$\hat{y} = 7.63604 + 0.11063A + 0.08812B + 0.05188D - 0.12979E - 0.08271B*E + 0.04229A*E$$

$$8.0 = 7.63604 + 0.11063(-1) + 0.08812(1) + 0.05188D - 0.12979(-1) - 0.08271(1)*(-1) + 0.04229(-1)*(-1)$$

$$\Rightarrow 8.0 = 7.86832 + 0.05188D \Rightarrow D = \frac{8.0 - 7.86832}{0.05188} = 2.54$$

In general, the **resolution** of a two-level fractional factorial design is equal to the **number of letters** in the **shortest word** in the **defining relation**.

The half of a 2^3 factorial with $ABC = I$ is called a **Resolution III design**, since there are 3 letters in the generating word. We write it as $\mathbf{2_{III}^{3-1}}$.

A half of a 2^4 factoral with defining relationship $ABC = I$ would also be resolution III, and would have $\mathrm{D} = ABCD$ but $A = BC$, etc.

If the defining relationship were $ABCD = I$ (**Resolution IV**), then $A = BCD$, $AB = CD$,etc. – main effects are aliased only with 3 factor interactions, 2 factor interactions with each other.

The larger the resolution, the better.

Obviously, the best we can do by running half of a 2^k design is Resolution K, with $AB \ldots K = I$.

One-quarter Fraction of Factorial Designs:

Consider a situation in which **six factors** (A,B,C,D,E, and F), each at **two levels**, are of interest, but the experimenters **cannot** afford to run all $2^6 = 64$ treatment combinations. They can, however, afford **sixteen** runs. This suggests a **one-quarter fraction** of a design.

Suppose we select the following two defining relationships

$$ABCE = I \text{ and } BCDF = I$$

This implies, for example,

$$I = ABCE * BCDF = AB^2C^2DEF = ADEF$$

$$I = ABCE \rightarrow E = ABCE^2 \rightarrow E = ABC$$

$$\mathrm{I} = \mathrm{BCDF} \rightarrow F = BCDF^2 \rightarrow F = BCD$$

$$\mathrm{I} = ADEF \rightarrow F = ADEF^2 \rightarrow F = ADE$$

One-quarter Fraction of Factorial Designs:

Consider a situation in which **six factors** (A,B,C,D,E, and F), each at **two levels**, are of interest, but the experimenters **cannot** afford to run all $2^6 = 64$ treatment combinations. They can, however, afford **sixteen** runs. This suggests a **one-quarter fraction** of a design. Suppose we select the following two defining relationships $ABCE = I$ and $BCDF = I$. This implies, for example, $E = ABC$ and $F = BCD$.

We construct the design by writing down all 16 rows of ± signs for a full 2^4 factorial in factors A, B, C and D. Then the signs for E and F are computed from $E = ABC$ and $F = BCD$.

This is a **Resolution IV design**: 2_{IV}^{6-2} and all two factor interactions are aliased with other two (or more) factor interactions. For instance

$\mathrm{I} = ABCE \rightarrow AB = CE,\ I = BCDF \rightarrow AB = ACDF$. So $\boldsymbol{AB = CE = ACDF}$

$\mathrm{I} = ABCE \rightarrow EF = ABCF,\ I = BCDF \rightarrow EF = BCDE$. So

$$\boldsymbol{EF = ABCF = BCDE}$$

There are more than one alternative when selecting the generators for a 2^{k-p} design. For example to generate a quarter fraction of a 2^6 design the generators $\boldsymbol{E = ABC}$ **and** $\boldsymbol{F = ABD}$ could be used, or the generators $\boldsymbol{E = AB}$ **and** $\boldsymbol{F = ACD}$ could be used.

The selections result in the defining relation and alias structure (for the main effects only) shown below:

First Selection	Second Selection
I = ABCE = ABDF = CDEF	I = ABE = ACDF = BCDEF
A + BCE + BDF + ACDEF	A + BE + CDF + ABCDEF
B + ACE + ADF + BCDEF	B + AE +ABCDF + CDEF
C + ABE + ABCDF + DEF	C + ABCE + ADF + BDEF
D + ABCDE + ABF + CEF	D + ABDE + ACF + BCEF
E + ABC + ABDEF + CDF	E + AB + ACDEF + BCDF
F + ABCEF + ABD + CDE	F + ABEF + ACD + BCDE

Question: Both generators results in 16-run fractional factorials, but which set of generators might be preferable?

There are more than one alternative when selecting the generators for a 2^{k-p} design. For example to generate a quarter fraction of a 2^6 design the generators $E = ABC$ and $F = ABD$ could be used, or the generators $E = AB$ and $F = ACD$ could be used. The selections result in the defining relation and alias structure (for the main effects only) shown below:

First Selection	Second Selection
I = ABCE = ABDF = CDEF	I = ABE = ACDF = BCDEF
A + BCE + BDF + ACDEF	A + BE + CDF + ABCDEF
B + ACE + ADF + BCDEF	B + AE +ABCDF + CDEF
C + ABE + ABCDF + DEF	C + ABCE + ADF + BDEF
D + ABCDE + ABF + CEF	D + ABDE + ACF + BCEF
E + ABC + ABDEF + CDF	E + AB + ACDEF + BCDF
F + ABCEF + ABD + CDE	F + ABEF + ACD + BCDE

The **first set** of generators might be **preferable**, since each main effect is confounded with three-factor interactions and higher order interactions, while the second set of generators results in a design where main effect A and B are confounded with one two-factor interaction each.

Three general criteria have been proposed for guiding the choice among various possible sets of generators for any 2^{k-p} design.

These criteria are the **resolution criteria**, the **aberration criteria**, and the **clear effect criteria**.

Resolution Criteria: In a **resolution** $\boldsymbol{l}$ design **no effect** involving $\boldsymbol{i}$ **factors** is **aliased** with effects of order **less** than $\boldsymbol{l-i}$.

Therefore, the **larger the resolution**, the **better**.

Aberration Criteria: When two or more design, created with different set of generators, have the same number of runs and the same resolution minimum aberration criteria can be used for deciding which design is preferable (See Fries, A. and Hunter, W. G. (1980), "**Minimum aberration 2^{k-p} design**", Technometrics, **22**, 601-608).

Clear Effect Criteria: An effect is clear if **none** of its aliases are main effects or two-factor interactions.

The **larger clear effects**, the **better**.

In some cases, a design that is not the minimum aberration design may have more clear effect than the minimum aberration design.

The easiest way to create a 2^{k-p} fractional factorial in R is to use the function **FrF2** in R package FrF2. The following R codes creates 16-runs 2^{5-1}design with generators $E = ABCD$.

```
> library(FrF2)
> design = FrF2(16,5, generators ="ABCD", randomize =FALSE)
> design
```

The following R codes creates 16-runs 2^{5-1}design with generators $E = ABCD$.

```
> library(FrF2)
> design = FrF2(16,5, generators ="ABCD", randomize =FALSE)
> design
```

```
     A  B  C  D  E
1   -1 -1 -1 -1  1
2    1 -1 -1 -1 -1
3   -1  1 -1 -1 -1
4    1  1 -1 -1  1
5   -1 -1  1 -1 -1
6    1 -1  1 -1  1
7   -1  1  1 -1  1
8    1  1  1 -1 -1
9   -1 -1 -1  1 -1
10   1 -1 -1  1  1
11  -1  1 -1  1  1
12   1  1 -1  1 -1
13  -1 -1  1  1  1
14   1 -1  1  1 -1
15  -1  1  1  1 -1
16   1  1  1  1  1
class=design, type= FrF2.generators
```

If the user does not specify generators when calling FrF2, the function automatically selects the set of generators that will result in a minimum aberration design. The following codes creates the minimum aberration 2^{8-4} design.

```
> library(FrF2)
> design = FrF2(16,8)
> design
```

	A	B	C	D	E	F	G	H
1	1	1	-1	-1	-1	-1	1	1
2	1	-1	1	1	-1	-1	1	-1
3	-1	-1	-1	-1	-1	-1	-1	-1
4	-1	1	1	-1	-1	1	1	-1
5	-1	1	1	1	-1	-1	-1	1
6	1	1	1	1	1	1	1	1
7	-1	1	-1	-1	1	1	-1	1
8	1	1	1	-1	1	-1	-1	-1
9	-1	-1	-1	1	-1	1	1	1
10	-1	-1	1	-1	1	-1	1	1
11	1	-1	-1	1	1	-1	-1	1
12	1	-1	1	-1	-1	1	-1	1
13	1	-1	-1	-1	1	1	1	-1
14	-1	1	-1	1	1	-1	1	-1
15	1	1	-1	1	-1	1	-1	-1
16	-1	-1	1	1	1	1	-1	-1

class=design, type= FrF2

The following codes creates the minimum aberration 2^{8-4} design.

```
> library(FrF2)
> design = FrF2(16,8)
> design
```

The **generators** function shows the generators for the design FrF2 created.

```
> library(DoE.base)
> generators(design)
```

$generators

[1] "E=ABC" "F=ABD" "G=ACD" "H=BCD"

The following codes creates the minimum aberration 2^{8-4} design.

```
> library(FrF2)   > design = FrF2(16,8)   > design
```

The **aliases** functions show the alias pattern for the design FrF2 created.

```
> y = runif(16,0,1)
> library(DoE.base)
> aliases(lm(y~(.)^3,
        data = design))
```

```
A = B:C:E = B:D:F = B:G:H = C:D:G = C:F:H = D:E:H = E:F:G
B = A:C:E = A:D:F = A:G:H = C:D:H = C:F:G = D:E:G = E:F:H
C = A:B:E = A:D:G = A:F:H = B:D:H = B:F:G = D:E:F = E:G:H
D = A:B:F = A:C:G = A:E:H = B:C:H = B:E:G = C:E:F = F:G:H
E = A:B:C = A:D:H = A:F:G = B:D:G = B:F:H = C:D:F = C:G:H
F = A:B:D = A:C:H = A:E:G = B:C:G = B:E:H = C:D:E = D:G:H
G = A:B:H = A:C:D = A:E:F = B:C:F = B:D:E = C:E:H = D:F:H
H = A:B:G = A:C:F = A:D:E = B:C:D = B:E:F = C:E:G = D:F:G
A:B = C:E = D:F = G:H
A:C = B:E = D:G = F:H
A:D = B:F = C:G = E:H
A:E = B:C = D:H = F:G
A:F = B:D = C:H = E:G
A:G = B:H = C:D = E:F
A:H = B:G = C:F = D:E
```

FrF2 can also create designs with the maximum number of clear effects.

For example, the call **FrF2(32,9)** produces the **minimum aberration 2^{9-4} design** that has **nine** clear main effects and **eight** clear two-factor interactions.

The call FrF2(32,9, MaxC2=TRUE) produces a **2^{9-4} design** that has all nine main effects and 15 clear two-factor interactions.

Some textbooks give tables for 2^{k-p} fractional factorial designs along with their generators and their clear effects. See, for example, Appendix 5A on pages 252 to 264 of "Experiments, Planning, Analysis, and Optimization", by Wu, C. F. J., and Hamada, S., Second edition, Wiley.

Example 2: Experiment were conducted to find the **optimal** conditions for culturing Paecilomyces variotii (a fungus commonly found in the air and soils of tropical countries) on eucalyptus hemicellulosic hydrolyzate with a view to producing microbial protein (See AlmeidaSilva, J. B., Lima, U. A., Taqueda, M. E. S., and Guaragna, F. G. (1998), "*Use of fractional factorial design for selection of nutrients for culturing Paecilomyces variotii in eucalyptus hemicellulosic hydrolysate*", Brazilian Journal of Chemical Engineering, **15**, 273-279).

Only 51.6% of the total dry mass of eucalyptus wood is utilized by Brazilian industry while the rest (branches, leaces, small trees, etc.) is left in the field.

The hemicellulose fraction of this waste can be easily removed by acid treatment, and the resulting hydrolyzate is rich in fermentable sugars.

The fungus P, variotii was selected from among 21 species of yeasts and filamentous for its performance on eucalyptus hemicellulose hydrolyzate.

Protein biomass produced by this fungus during 72 hours of fermentation has an amino acid profile that is equal to exceeds conventional products used for animal feet.

The **purpose** of the experiments was to study the **influence** of inhibitors, nutrients, and fermentation time on the biomass produced by P, variotii.

A 2^{8-4} fractional factorial design was used with following generators:

$$E = BCD, F = ACD, G = ABC, \text{and } H = ABD.$$

The following tables shows the factors and levels used in the study.

Label	Factor	Levels	
		−	+
A	Inhibitors (Furfural and Acetic Acid)	1.25g/L	7.8g/L
B	Rice Bran	10.0g/L	30.0g/L
C	Urea	0.0g/L	2.0g/L
D	Magnesium Sulfate	0.0g/L	1.5g/L
E	Ammonium Sulfate	0.0g/L	2.0g/L
F	Potassium Nitrate	0.0g/L	2.0g/L
G	Sodium Phosphate	0.0g/L	2.0g/L
H	Fermentation Time	72 hrs.	96 hrs.

A 2^{8-4} fractional factorial design was used with generators:

$$E = BCD, F = ACD, G = ABC, \text{and } H = ABD.$$

This is the **minimum aberration resolution IV** design, and the **clear effect** in this design are the **eight main effects**.

Example 2 (Cont.): The experimental design that was used was a half fraction of a 2^{8-4} full factorial design. The design matrix and response data are given in the following table:

Factor								
A	B	C	D	E	F	G	H	y
−	−	−	−	−	−	−	−	5.75
+	−	−	−	−	+	+	+	6.70
−	+	−	−	+	−	+	+	11.12
+	+	−	−	+	+	−	−	10.67
−	−	+	−	+	+	+	−	4.92
+	−	+	−	+	−	−	+	5.35
−	+	+	−	−	+	−	+	2.81
+	+	+	−	−	−	+	−	10.83
−	−	−	+	+	+	−	+	6.08
+	−	−	+	+	−	+	−	7.27
−	+	−	+	−	+	+	−	9.68
+	+	−	+	−	−	−	+	4.20
−	−	+	+	−	−	+	+	3.90
+	−	+	+	−	+	−	−	3.78
−	+	+	+	+	−	−	−	11.57
+	+	+	+	+	+	+	+	7.39

Example 2: A 2^{8-4} fractional factorial design was used with generators:

$$\boldsymbol{E = BCD, F = ACD, G = ABC, \text{and } H = ABD}.$$

There are 28 two factor interactions , but only seven of them are estimated, since they are confounded in strings of four interactions. The seven aliased strings of two-factor interactions that can be estimated are shown in the following:

$$CG + DH + AB + EF$$
$$AC + BG + DF + EH$$
$$CF + AD + EG + BH$$
$$CH + DG + AE + BF$$
$$CD + GH + AF + BE$$
$$BC + AG + DE + FH$$
$$CE + FG + AH + BD$$

Note that the **lm** function estimates all the two factor interactions with A, that is AB, AC, AD, AE, AF, AG, and AH.

R Codes to Enter Data:

```
> y= c( 5.75, 6.70,11.12,10.67, 4.92, 5.35, 2.81,10.83,
        6.08, 7.27, 9.68, 4.20, 3.90, 3.78,11.57, 7.39)

> A = rep(c(-1,1), times=8)
> B = rep(c(-1,1), each= 2, times=4)
> C = rep(c(-1,1), each= 4, times =2)
> D = rep(c(-1,1), each = 8)

> E = B*C*D
> F = A*C*D
> G = A*B*C
> H = A*B*D
```

R codes to Enter Data:

```
> A = factor(A)
> B = factor(B)
> C = factor(C)
> D = factor(D)
> E = factor(E)
> F = factor(F)
> G = factor(G)
> H = factor(H)
```

	A	B	C	D	E	F	G	H	y
1	-1	-1	-1	-1	-1	-1	-1	-1	5.75
2	1	-1	-1	-1	-1	1	1	1	6.70
3	-1	1	-1	-1	1	-1	1	1	11.12
4	1	1	-1	-1	1	1	-1	-1	10.67
5	-1	-1	1	-1	1	1	1	-1	4.92
6	1	-1	1	-1	1	-1	-1	1	5.35
7	-1	1	1	-1	-1	1	-1	1	2.81
8	1	1	1	-1	-1	-1	1	-1	10.83
9	-1	-1	-1	1	1	1	-1	1	6.08
10	1	-1	-1	1	1	-1	1	-1	7.27
11	-1	1	-1	1	-1	1	1	-1	9.68
12	1	1	-1	1	-1	-1	-1	1	4.20
13	-1	-1	1	1	-1	-1	1	1	3.90
14	1	-1	1	1	-1	1	-1	-1	3.78
15	-1	1	1	1	1	-1	-1	-1	11.57
16	1	1	1	1	1	1	1	1	7.39

```
> data = data.frame(A,B,C,D,E,F,G,H,y)
> data
```

The function FrF2 also can be used to create the design and data.

```
> library(FrF2)

> design = FrF2(16,
        generators =
        c("BCD", "ACD", "ABC", "ABD"),
                randomize = FALSE)

> y= c( 5.75, 6.70,11.12,10.67,
        4.92, 5.35, 2.81,10.83,
        6.08, 7.27, 9.68,  4.20,
        3.90, 3.78,11.57, 7.39)

> data = add.response(design,y)
> data
```

```
   A  B  C  D  E  F  G  H     y
1 -1 -1 -1 -1 -1 -1 -1 -1  5.75
2  1 -1 -1 -1 -1  1  1  1  6.70
3 -1  1 -1 -1  1 -1  1  1 11.12
4  1  1 -1 -1  1  1 -1 -1 10.67
5 -1 -1  1 -1  1  1  1 -1  4.92
6  1 -1  1 -1  1 -1 -1  1  5.35
7 -1  1  1 -1 -1  1 -1  1  2.81
8  1  1  1 -1 -1 -1  1 -1 10.83
9 -1 -1 -1  1  1  1 -1  1  6.08
10  1 -1 -1  1  1 -1  1 -1  7.27
11 -1  1 -1  1 -1  1  1 -1  9.68
12  1  1 -1  1 -1 -1 -1  1  4.20
13 -1 -1  1  1 -1 -1  1  1  3.90
14  1 -1  1  1 -1  1 -1 -1  3.78
15 -1  1  1  1  1 -1 -1 -1 11.57
16  1  1  1  1  1  1  1  1  7.39
class=design, type= FrF2.generators
```

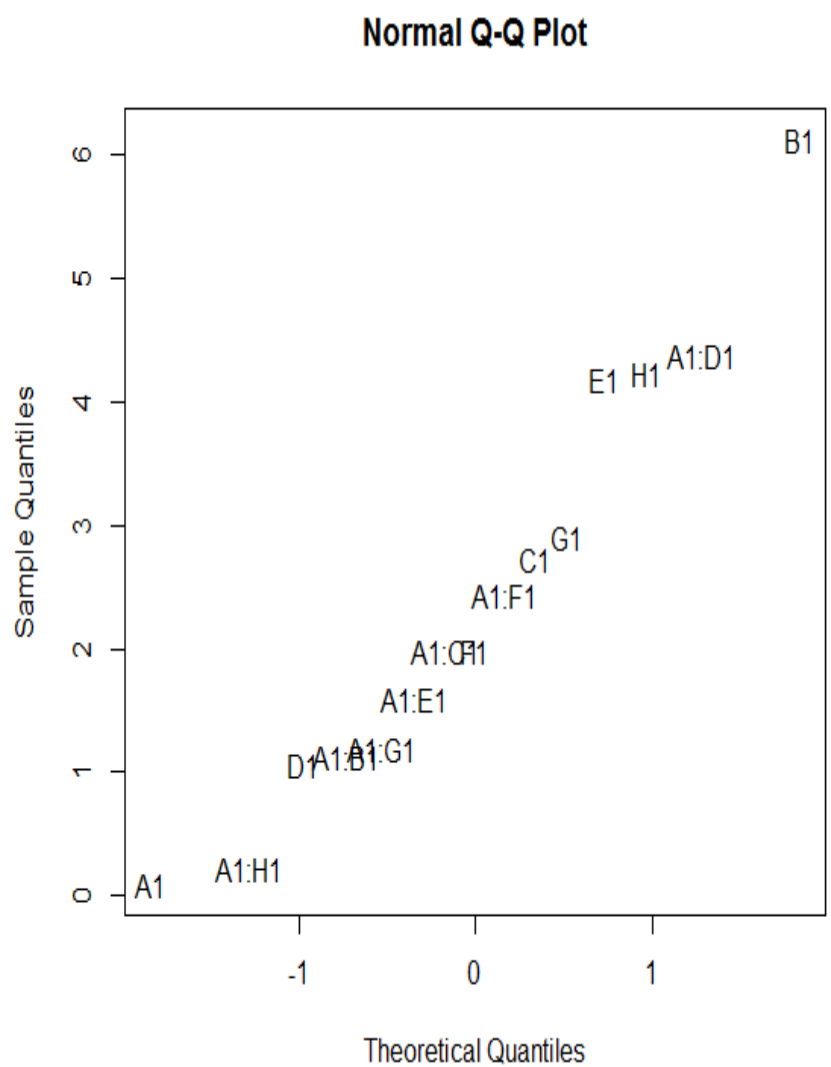
R Codes to Obtain ANOVA Table:

```
> model = lm(y~(.)^8, data = data)
> anova(model)
```

```
Analysis of Variance Table

Response: y
          Df Sum Sq Mean Sq F value Pr(>F)
A          1  0.008   0.008
B          1 37.577  37.577
C          1  7.453   7.453
D          1  1.145   1.145
E          1 17.472  17.472
F          1  3.960   3.960
G          1  8.410   8.410
H          1 17.893  17.893
A:B        1  1.288   1.288
A:C        1  3.940   3.940
A:D        1 19.228  19.228
A:E        1  2.544   2.544
A:F        1  5.929   5.929
A:G        1  1.428   1.428
A:H        1  0.051   0.051
Residuals  0  0.000
Warning message:
In anova.lm(model) :
  ANOVA F-tests on an essentially perfect fit are unreliable
```

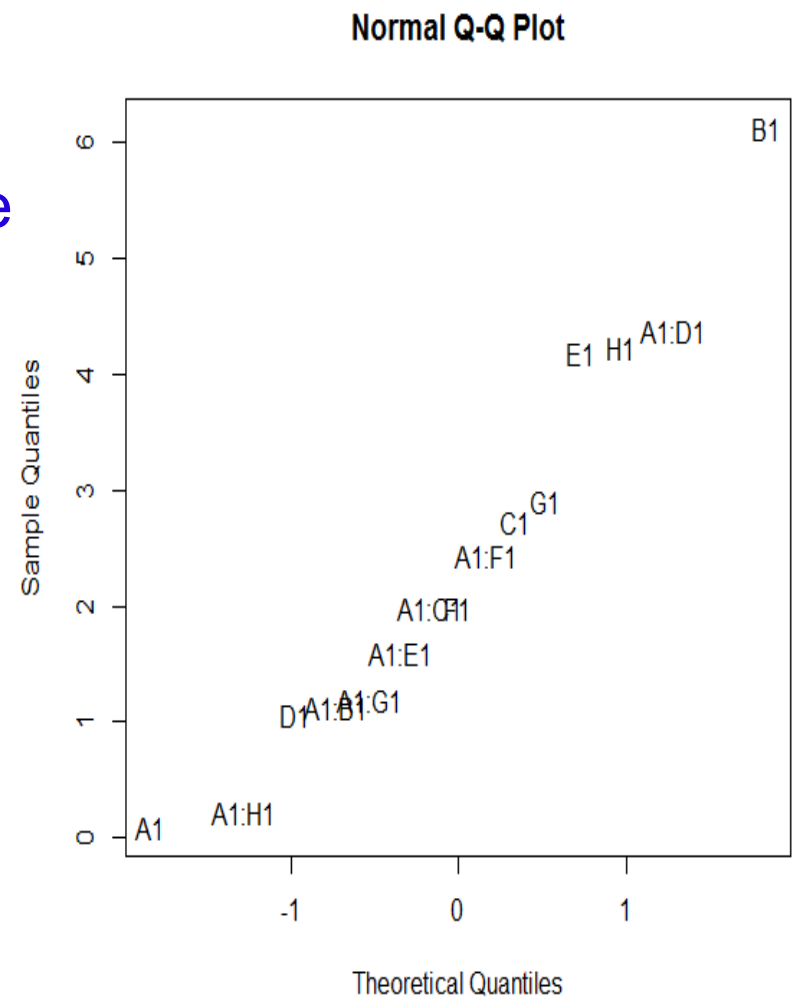
R Codes to Obtain Half-Normal Plot:

```
> model = lm(y~(.)^8, data = data)
> anova(model)

> model$effects
> model.effects =
        abs(model$effects[-1])


> qq = qqnorm(model.effects,
                        type="n")
>text(qq$x, qq$y, labels =
          names(model.effects))
```



Half-Normal Plot:

Main effects for B, E, and H and Interaction AD are **largest** in absolute Value.



If the **three-factor interactions** are assumed to be **negligible**, the effect For the three main effects could be Interpreted as the cause of the Largest.

But it is **not** clear what the confounded interaction $\mathrm{CF} + AD + EG + BH$ represents.

The authors of article decided to run **another** resolution V 2^{5-1} follow-up experiment using factor B, C, E , G and H with relation $BCEGH = I$ and factors A, D, and F held constant at the mid-point of the levels used in the first experiment.

See the "*Design and Analysis of Experiment With R*", by J. Lawson for detailed analysis of follow-up design.

What Have We Learned?

We discussed fractional factorial design, specially one-half and one-quarter fractional factorial design in this chapter.